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Original Report

A choice of radionuclide: Comparative outcomes and toxicity of ruthenium-106 and iodine-125 in the definitive treatment of uveal melanoma

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Abstract

Purpose: Both iodine-125 (^{125}I) Collaborative Ocular Melanoma Study and ruthenium-106 (^{106}Ru) eye plaques can achieve excellent tumor control in patients diagnosed with uveal melanoma. We analyzed our single institutional experience in the management of ocular melanoma treated with either ^{125}I or ^{106}Ru plaque brachytherapy.

Methods and materials: The records of 107 patients with uveal melanoma treated with either ^{106}Ru (n = 40) or ^{125}I (n = 67) plaque brachytherapy between 2000 and 2008 were retrospectively reviewed. Tumor control parameters and toxicity were assessed.

Results: Actuarial 5-year rates of local control, progression-free survival, and overall survival with ^{106}Ru were 97%, 94%, and 92%, respectively. For ^{125}I, these values were 83%, 65%, and 80%. In the subset of patients with tumor apex height $\leq$5 mm (36 ^{125}I and 40 ^{106}Ru), there was no difference in overall survival; however, progression-free survival was significantly improved with ^{106}Ru (P = .02). Enucleation-free survival was significantly different between the 2 subsets, with no enucleations in the ^{106}Ru cohort (P = .02). Patients treated with ^{106}Ru experienced reduced retinopathy (P = .03) and cataracts (P < .01).

Conclusions: Both ^{125}I and ^{106}Ru eye plaque brachytherapy treatment result in encouraging tumor control for patients with uveal melanoma. We demonstrate that ^{106}Ru offers these benefits with reduced toxicity in patients treated for uveal melanomas $\leq$5 mm in apical height.

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Conflicts of interest: None.
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